JUDGES' LIBRARY / DOSSIER 01

The mission & the demo.

A concise story for the room: what PROHOR addresses, what a judge can try, and where the evidence begins.

The idea in one sentence. PROHOR helps a crew member turn separate health readings into an explainable, personal comparison and a human-review handover — even when a previously cached app is offline.

The operational gap

A long-duration crew needs to track health while dealing with limited resources and potential communication delays. Numbers alone can be hard to interpret: a change has meaning only when the person, measurement method, prior observations and missing data are visible.

PROHOR is a research prototype, not a medical service. It brings the observation, baseline explanation, research context and handover into one interface. It does not diagnose illness, set flight rules or replace a clinician.

What makes the experience inspectable

Moment	What the interface shows	Why it matters
Observe	A reading, its unit and a labeled data mode.	Separates synthetic demonstration from personal input.
Compare	A prior-day median, band and coverage.	A reviewer can inspect the rule instead of trusting a score.
Review	Research context and a structured handover.	Keeps the next decision with a human.

Designed for the 2026 NASA Space Apps health-monitoring challenge. Independent team project; no NASA endorsement is implied.

02 / GUIDED WALKTHROUGH

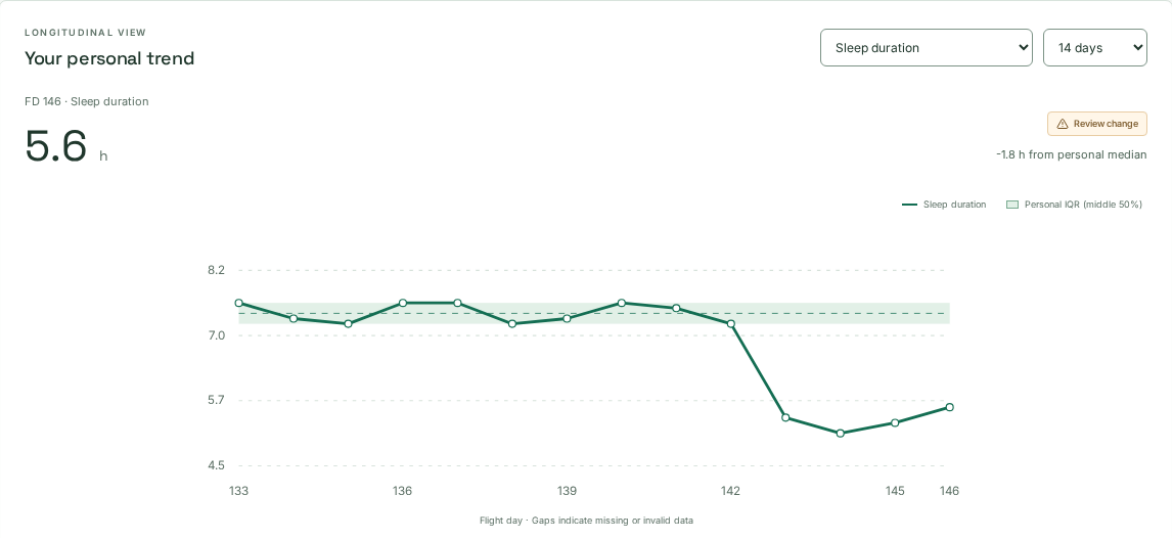
A demo judges can reproduce.

The following steps use the built-in **synthetic** CDR-01 scenario. Nothing in this walkthrough is a real astronaut's personal health record.

Step	Action	What to point out
1 Start	Open Mission overview and select "See the change" in the guided tour.	The tour selects demo crew CDR-01 on flight day 146 and opens Trends.
2 Explain	Inspect sleep in Trends & baselines.	5.6 h is simulated; the 7.4 h personal median is from the synthetic enrollment window. The flagged state is a prototype rule, not a medical finding.
3 Investigate	Use "Explore the evidence" from the overview tour.	The NASA explorer is a separate historical Inspiration4 study. Open its source and choose analyte and subject.
4 Hand over	Use "Prepare for review" to open Crew handover.	Inspect a local export and the explicit statement that nothing was transmitted. A human decides what comes next.
5 Explore	Open Data & settings for local CSV tools.	A fresh local profile must build its own baseline; importing data is not the same as proving device integration.

Offline check: Visit online first, wait for the service worker and cached assets, then turn off the network and reload. Cache eviction, browser restrictions or a failed first visit can prevent offline operation.

A well-framed presentation starts with the problem, demonstrates one interaction all the way through, then closes with limitations. The evidence dossiers provide deeper technical detail when a judge asks.



Actual PROHOR Trends view. CDR-01 and all readings shown are synthetic.

03 / POSITIONING

The promise, honestly bounded.

Who benefits and how

The intended users are crew members and their reviewers in a future mission-health workflow. Today, the prototype demonstrates a useful interaction pattern for transparent self-observation. It has not been evaluated with an operational crew or shown to improve medical outcomes.

What a judge can verify

Question	Where to look
Does the demo use real or invented readings?	Demo is explicitly synthetic; user entries are local; NASA research is separately labeled.
Can I verify the scientific source?	NASA evidence links the OSD-575 archive and verifies the bundled submitted CSV against its manifest.
Why was a change flagged?	Trends exposes the baseline window, median, day coverage, directional rule and change band.
Does this replace an expert?	No. Routine prompts and handover support human review; symptoms and approved procedures take precedence.

Team

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Sources & further reading

- Live prototype
- Official challenge
- Repository and reproducible source